

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Hospital information system for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS with fields of patient name, age, address, phone, email, Attender name, attender phone, doctor name, diseases, etc.

Experiment 2: Create a simple cloud software application for online super market using any Cloud Service Provider to demonstrate SaaS with the template of customer name, address, phone, email, items, quantity etc. with the item classification form.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 3 CPU, 10GB RAM and 10GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

Assessment Tasks

Experiment 1: Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as name of the student, Reg no of the student , Subjects , Marks, Average, percentage, rank, overall performance etc.

Experiment 2: Create a simple cloud software application for Property Buying & Rental process (In Chennai city) using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Buy, Rent, commercial, Rental Agreement , Rental Agreement, Area, place, Range, property value etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 5 CPU, 12 GB RAM and 8 GB storage disk using a Type 2 Virtualization Software.

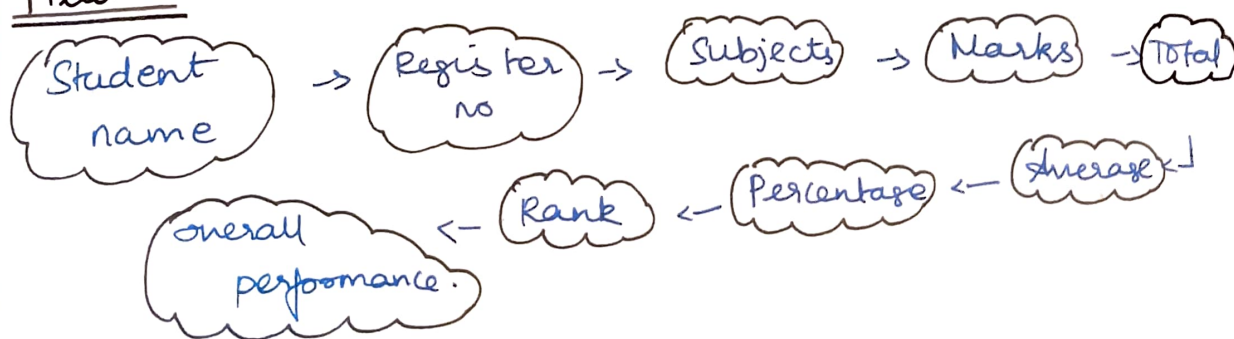
Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

EXPERIMENT-1Cloud Based Mark sheet processing system.Aim:

To create a simple Cloud Based Mark sheet processing system using a cloud Service provider to demonstrate (SaaS).

Objective : Requirements :

- ⇒ Total ⇒ Computer / Laptop
⇒ Internet connection
⇒ Web browser
⇒ HTML, CSS, JSS
⇒ Firebase cloud service.

Fields:procedure :

1. Create a web app
2. Create a form to enter student name & register no.

3. Enter marks for different subjects.
4. Calculate total marks.
5. Calculate the average & percentage.
6. Calculate the overall performance.
7. Store the student details in cloud.

Total marks } = Sum of all subject marks

Avg : Total marks / No. of subjects.

percentage = (Total marks) / (Max mark) $\times 100$.

Input:

Vijay

101

80

90

85

75

70

output:

Vijay

101

400

80

80%.

very good.

Result:

Thus, Cloud Based Mark sheet processing system was successfully created & deployed using a cloud service, demonstrating SaaS.

EXPERIMENT-2

Cloud Based property Buying & Rental System.

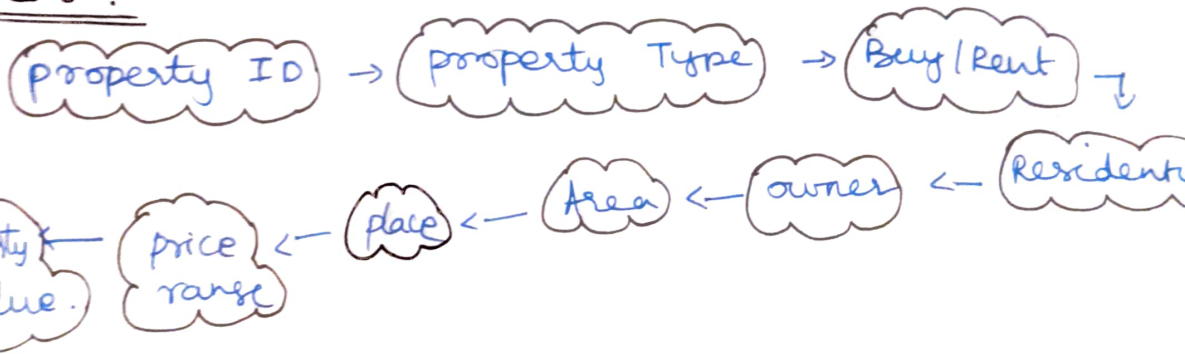
Aim:

To create a simple cloud-based property Buying & rental system for Chennai city using a cloud service provider to demonstrate SaaS.

Requirements

- ⇒ Computer / Laptop
- ⇒ Internet Connection
- ⇒ Web browser
- ⇒ Firebase

Fields:



Procedure:

1. Create a web application.
2. Create a form to enter property fields.
3. Enter the property type & location.
4. Select Buy & rent
5. Select Residential / Commercial
6. Enter area & property value.

EXPERIMENT-2

Cloud Based property Buying & Rental System.

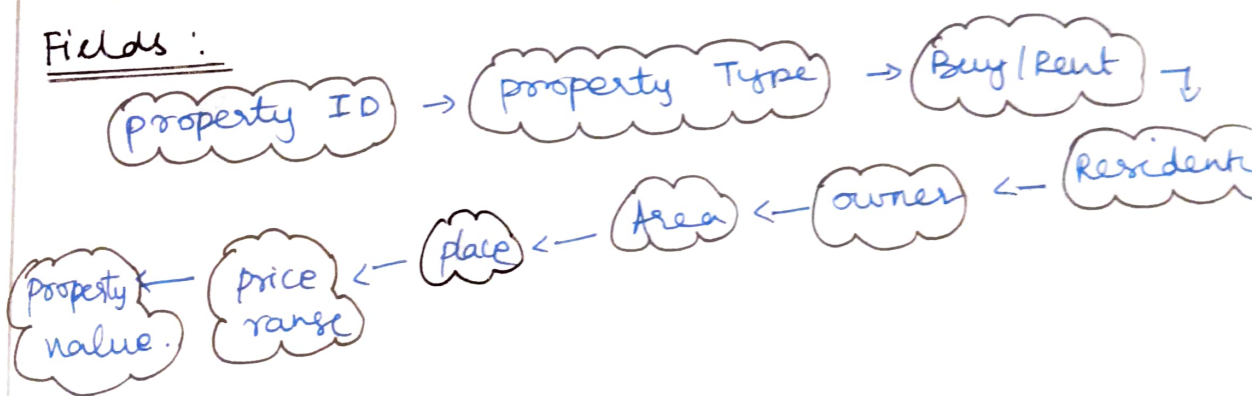
Aim:

To create a simple cloud-based property buying & rental system for Chennai city using a cloud service provider to demonstrate SaaS.

Requirements

- ⇒ Computer / Laptop
- ⇒ Internet Connection
- ⇒ Web browser
- ⇒ Firebase

Fields:



Procedure:

1. Create a web application.
2. Create a form to enter property fields.
3. Enter the property type & location.
4. Select Buy & rent
5. Select residential / Commercial
6. Enter area & property value.

7. Enter rental agreement detail
8. Store the property information in the cloud.
9. display available properties
10. Test the app.

Input:

P101
Flat
Rent
residence
1200 Sq.ft
Chennai
₹ 20 000/month
₹ 60 00 000

Output

P101
Apartment
Chennai
1200 Sq.ft
Available
₹ 20,000/month

Result:

Thus, the cloud-Based property buying & rental system was successfully created & deployed using a cloud service, demonstrating

EXPERIMENT - 3

Creation of Virtual Machine Type - 2 Hypervisor.

Aim :

To demonstrate Virtualization by installing a Type 2 Hypervisor and Creating a VM.

Requirements :

- ⇒ Computer .
- ⇒ Window .
- ⇒ Oracle .
- ⇒ Windows .

Procedure :

1. Download the Oracle VirtualBox.
2. open Oracle Virtual Box .
3. Click New to Create a virtual machine.
4. Enter the name of the virtual machine.
5. Select the required OS.
6. allocate 5 CPU'S to VM .
7. Allocate 12 GB RAM .
8. Create a virtual hard disk of 80GB.

Configuration :

Marksheet VM

5 CPUs

12 GB

8 GB

Oracle Virtual Box.

Windows / Linux.

Result:

Thus, a virtual machine with 5 CPU, 12 GB ram & 8 GB storage was successfully

Experiment - 4.

Snapshot & Restoration of virtual machine:

Aim:

to create a Snapshot of a Virtual machine and restore the previous version of the virtual machine.

Requirements

↳ Oracle Virtual Box

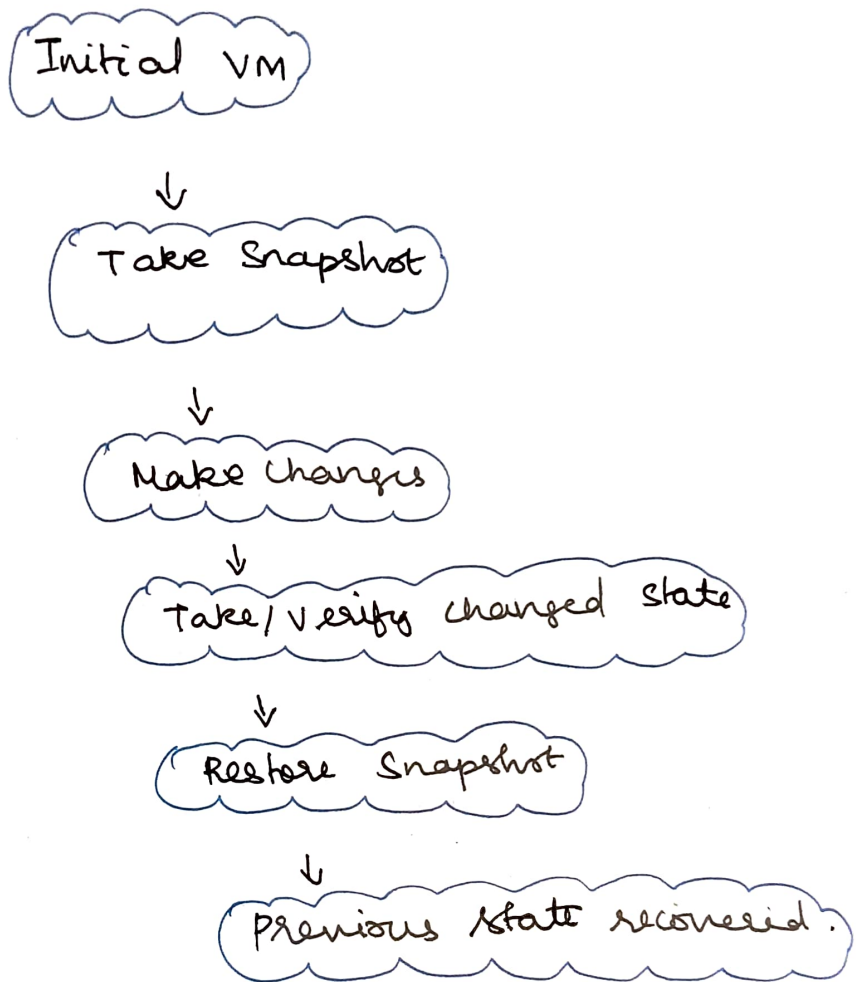
↳ previously created Virtual machine.

procedure:

1. open Virtual Box.
2. Select the previously created virtual machine.
3. Start the virtual machine.
4. Make some changes to the VM.
5. Shut down the VM.
6. Click take Snapshot.
8. Enter a suitable snapshot name such as Initial state.

9. Start the VM
10. make some additional changes.
11. Shutdown VM

Eg:



Result:

Thus, a snapshot of the virtual machine was successfully created & the previous version was restored successfully using Oracle Virtual Box.